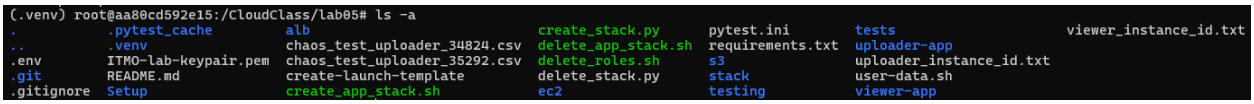
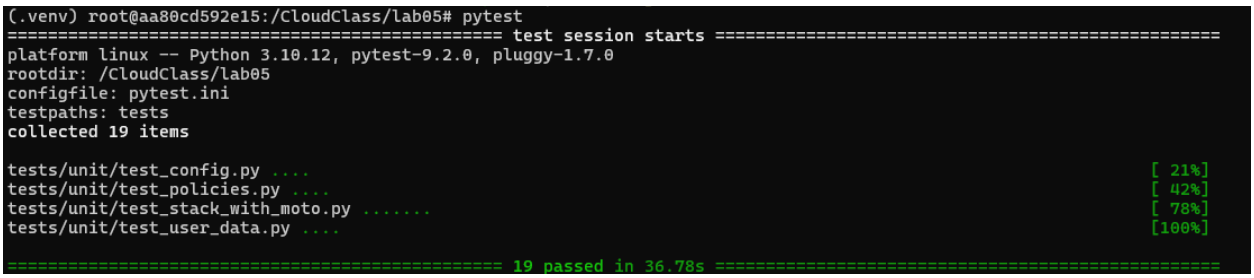
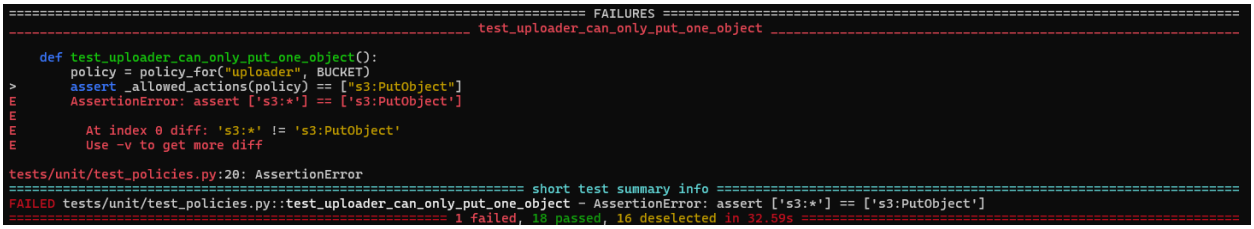
Output of ls -a lab05 (or a screenshot) showing the layout from Part 0 and section 4.



Screenshot of pytest showing 19 passed.



Screenshot of one "break it on purpose" experiment: the changed line and the failing test message.



Screenshot of python create_stack.py completing, with both URLs.

```
(.venv) root@aa80cd592e15:/CloudClass/lab05# python create_stack.py
Found credentials in shared credentials file: ~/.aws/credentials
== S3 bucket ==
Bucket itmo-gcm131-fileshare already exists, skipping.
== IAM roles and instance profiles ==
uploader role and instance profile are ready.
viewer role and instance profile are ready.
Waiting 15s for IAM propagation...
== Launch templates ==
Template itmo-uploader-lt exists, published version 13
Template itmo-viewer-lt exists, published version 13
== Network ==
Created ALB security group sg-086e45725930904be
== Target groups, load balancer, listeners ==
Waiting for the ALB to become active...
== Auto Scaling Groups ==
Created Auto Scaling Group itmo-uploader-asg
Created Auto Scaling Group itmo-viewer-asg

Uploader: http://itmo-app-alb-1544934435.us-east-2.elb.amazonaws.com:8080/
Viewer:   http://itmo-app-alb-1544934435.us-east-2.elb.amazonaws.com:8081/
Give the target groups 1 to 2 minutes to report healthy, then run: pytest -m integration
```

Screenshot of pytest -m integration -v with all 14 tests passing.

```
(.venv) root@aa80cd592e15:/CloudClass/lab05# pytest -m integration -v
============================= test session starts ==============================
platform linux -- Python 3.10.12, pytest-9.2.0, pluggy-1.7.0 -- /CloudClass/lab05/.venv/bin/python3
cachedir: .pytest_cache
rootdir: /CloudClass/lab05
configfile: pytest.ini
testpaths: tests
collected 35 items / 21 deselected / 14 selected

tests/integration/test_application.py::test_uploader_serves_its_form PASSED [  7%]
tests/integration/test_application.py::test_viewer_serves_its_page PASSED [ 14%]
tests/integration/test_application.py::test_text_file_uploaded_on_one_tier_appears_on_the_other PASSED [ 21%]
tests/integration/test_application.py::test_non_txt_file_is_rejected PASSED [ 28%]
tests/integration/test_application.py::test_file_over_1mb_is_rejected PASSED [ 35%]
tests/integration/test_infrastructure.py::test_bucket_blocks_all_public_access PASSED [ 42%]
tests/integration/test_infrastructure.py::test_uploader_role_can_only_write_the_shared_object PASSED [ 50%]
tests/integration/test_infrastructure.py::test_viewer_role_can_read_but_not_write PASSED [ 57%]
tests/integration/test_infrastructure.py::test_load_balancer_is_public_and_spans_two_azs PASSED [ 64%]
tests/integration/test_infrastructure.py::test_listeners_forward_to_the_right_target_groups PASSED [ 71%]
tests/integration/test_infrastructure.py::test_target_groups_use_fast_health_checks PASSED [ 78%]
tests/integration/test_infrastructure.py::test_auto_scaling_groups_are_configured_for_recovery PASSED [ 85%]
tests/integration/test_infrastructure.py::test_app_port_is_reachable_only_from_the_load_balancer PASSED [ 92%]
tests/integration/test_infrastructure.py::test_every_target_becomes_healthy PASSED [100%]

======================= 14 passed, 21 deselected in 4.68s ======================
```

Output of pytest -m chaos -s for both tiers, and the same run failing after you tightened the SLO.

```
=========================== short test summary info ============================
FAILED tests/integration/test_recovery.py::test_tier_recovers_within_the_slo[uploader] - AssertionError: assert 82.79205513000488 <= 60
FAILED tests/integration/test_recovery.py::test_tier_recovers_within_the_slo[viewer] - AssertionError: assert 87.07394456863403 <= 60
================== 2 failed, 33 deselected in 171.94s (0:02:51) ================
```

Screenshot of python delete_stack.py completing, and a second run completing with no errors.

```
(.venv) root@aa80cd592e15:/CloudClass/lab05# python delete_stack.py
Found credentials in shared credentials file: ~/.aws/credentials
== Auto Scaling Groups (this also terminates their instances) ==
itmo-uploader-asg still deleting after 300s. Check the console.
itmo-viewer-asg still deleting after 300s. Check the console.
== Load balancer and target groups ==
Deleted load balancer itmo-app-alb
Traceback (most recent call last):
  File "/CloudClass/lab05/delete_stack.py", line 23, in <module>
    main()
  File "/CloudClass/lab05/delete_stack.py", line 20, in main
    delete_stack(settings, AwsClients.create(settings.region), keep_foundation=args.keep_foundation)
  File "/CloudClass/lab05/stack/runner.py", line 63, in delete_stack
    balancer.delete_target_group(aws, tier.tg_name)
  File "/CloudClass/lab05/stack/balancer.py", line 84, in delete_target_group
    aws.elbv2.delete_target_group(TargetGroupArn=arn)
  File "/CloudClass/lab05/.venv/lib/python3.10/site-packages/botocore/client.py", line 606, in _api_call
    return self._make_api_call(operation_name, kwargs)
  File "/CloudClass/lab05/.venv/lib/python3.10/site-packages/botocore/context.py", line 123, in wrapper
    return func(*args, **kwargs)
  File "/CloudClass/lab05/.venv/lib/python3.10/site-packages/botocore/client.py", line 1094, in _make_api_call
    raise error_class(parsed_response, operation_name)
botocore.errorfactory.ResourceInUseException: An error occurred (ResourceInUse) when calling the DeleteTargetGroup operation: Target group 'arn:aws:elasticloadb
alancing:us-east-2:585335547635:targetgroup/itmo-uploader-tg/2083202918ad7558' is currently in use by a listener or a rule
(.venv) root@aa80cd592e15:/CloudClass/lab05# python delete_stack.py
Found credentials in shared credentials file: ~/.aws/credentials
== Auto Scaling Groups (this also terminates their instances) ==
== Load balancer and target groups ==
Deleted target group itmo-uploader-tg
Deleted target group itmo-viewer-tg
== Security groups ==
Deleted ALB security group sg-0e83607e08cf09d30
== Launch templates, IAM, bucket ==
Deleted launch template itmo-uploader-lt
Removed uploader role and instance profile.
Deleted launch template itmo-viewer-lt
Removed viewer role and instance profile.
Deleted bucket itmo-gcm131-fileshare
Cleanup complete.
```

One paragraph on what moto proved, what it could not, and which tests cover the gap.

Moto proved that the python code sends the right commands to AWS in the right order, that it creates resources properly and cleans them up properly. Because it is a fake AWS that runs in memory it cannot properly test if the real aws servers are working like if the instance installs node.js successfully or if the server boots up. The integration and chaos tests cover this gap by testing the real deployed AWS stack to ensure that the live application works, target group health checks, IAM policy permissions align and that the recovery happens in the time limit.

One paragraph comparing ignore_errors("EntityAlreadyExists") with 2>/dev/null || true

ignore_errors("EntityAlreadyExists") gives the user the ability to target certain errors giving them very fine control over what errors are overlooked and ignored and which are let through to print. This is very helpful in the case of typos and other unexpected failures. On the other hand **2>/dev/null || true** does not give the user any control to target certain error groups to block as it blocks all of them. This can become an issue because while its intended to block expected failures if an unexpected failure shows up then you will have a much harder time debugging and fixing the error.

Your answer to Challenge 5.

The S3 bucket creation was shorter in bash (create_app_stack.sh) however it is much longer in python(storage.py). One of the benefits is explicit error handling, which allows distinction between errors which allows silent failures to be shown and lets its tests with moto successfully output errors prior to ever touching the cloud letting you ensure that what is being run is correct.